

Bidirectional Sign Language Communication System

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Abstract—Effective communication between deaf and hearing communities is often impeded by language barriers, necessitating innovative solutions. Our project introduces a real-time sign language translation application designed to convert sign language gestures into audible speech, thereby facilitating seamless interaction. The system employs advanced computer vision and machine learning techniques to capture and interpret hand gestures. A camera records the user's hand movements, which are processed by a trained neural network to recognize specific gestures. These gestures are then mapped to corresponding words or phrases and synthesized into audible speech, enabling effective communication between deaf and hearing individuals. Preliminary evaluations demonstrate promising accuracy in both gesture recognition and speech synthesis, highlighting the system's potential across various settings. This technology not only promotes inclusivity and accessibility for individuals with hearing impairments but also offers adaptability to multiple sign languages, broadening its applicability. Future developments may include expanding the gesture database, enhancing recognition accuracy, and incorporating features such as real-time feedback to further improve communication effectiveness. This real-time sign language translation application represents a significant advancement in bridging the communication divide between deaf and hearing communities, fostering a more inclusive society.

Keywords—*Gesture Recognition, Computer Vision, Deep Learning, MLP Classifier, Hand Landmarks, Real-Time Processing.*

I. INTRODUCTION

Effective communication is fundamental to human interaction, yet millions of individuals with hearing and speech impairments face significant barriers due to the reliance on spoken language. Sign language serves as a vital mode of communication for these communities, enabling them to convey thoughts and emotions through structured gestures and expressions. However, the lack of widespread familiarity with sign languages among non-signers perpetuates social isolation and miscommunication [2]. According to the World Health Organization (WHO), over 63 million people in India suffer from auditory impairment, highlighting the urgent need for solutions tailored to Indian Sign Language (ISL).

While numerous research efforts have focused on sign language recognition systems, most have been limited to single-handed gestures used in American Sign Language (ASL) [2], [3]. Indian Sign Language (ISL), which relies on

both hands and includes unique gestures, remains underexplored due to its complexity and the absence of standardized datasets [1], [4]. Existing systems often suffer from high latency, limited gesture databases, dependency on expensive hardware like sensor gloves, and a lack of bidirectional translation capabilities [5], [6]. These limitations underscore the need for innovative approaches that integrate real-time processing, scalability, and accessibility.

This study introduces a bidirectional ISL communication system that bridges the communication gap between deaf and hearing communities. The proposed system enables real-time translation of ISL gestures into audible speech while also converting text and speech into animated ISL gestures. By leveraging computer vision techniques such as MediaPipe for hand tracking [9], [10] and deep learning models like Multilayer Perceptron (MLP) classifiers for gesture recognition [7], the system achieves robust performance without requiring specialized hardware beyond a webcam. Additionally, rule-based models are employed to generate sign animations from text and speech inputs, ensuring seamless interaction.

The contributions of this research are threefold: (1) development of a real-time ISL gesture recognition system using computer vision and deep learning; (2) implementation of text-to-sign and speech-to-sign functionalities for bidirectional communication; and (3) creation of an accessible platform that promotes inclusivity for individuals with hearing impairments. Preliminary evaluations demonstrate promising accuracy in gesture recognition, speech synthesis, and sign generation, showcasing the system's potential across diverse settings [7], [8], [10].

II. LITERATURE REVIEW

Several studies have explored sign language recognition using various methodologies, including computer vision, deep learning, wearable electronics, and artificial intelligence (AI), have been explored to advance sign language recognition systems. However, many of these approaches face challenges such as hardware dependency, limited vocabulary, computational overhead, and lack of bidirectional communication capabilities. Table 1 provides an overview of relevant research studies, highlighting their methodologies, results, advantages, and drawbacks while identifying research gaps that inform the development of a real-time bidirectional Indian Sign Language (ISL) communication system.

TABLE I. SUMMARY OF EXISTING RESEARCH ON SIGN LANGUAGE RECOGNITION

S.NO.	Title of Paper	Methodology	Results	Drawbacks
1.	Sign Language Recognition Using Convolutional Neural Networks (2015) [3]	Employed CNNs to recognize 20 Italian Sign Language gestures using Microsoft Kinect for depth images.	High accuracy in recognizing predefined gestures.	Limited to predefined gestures; reliance on Kinect hardware restricts accessibility.
2.	Indian Sign Language Recognition Using MediaPipe Holistic (2023) [10]	Utilized MediaPipe Holistic for hand, face, and pose landmarks; implemented LSTM networks for dynamic gestures.	Effective for static gestures (CNNs) and dynamic gestures (LSTMs).	Increased computational complexity due to multi-part processing; requires significant computational resources.
3.	Dynamic Sign Language Recognition Using MediaPipe Library and Modified LSTM Method (2023) [9]	Used MediaPipe Holistic keypoints with modified LSTM networks for static and dynamic gesture recognition.	Achieved 99.4% accuracy for dynamic sign language words; individual word accuracies reached up to 85%.	Requires high-quality video input; faces challenges with gesture variations across individuals.
4.	Recognition of Indian Sign Language (ISL) Using Deep Learning Model (2021) [7]	Used skin color segmentation and face elimination for hand tracking; SVM-based classification.	Improved real-time gesture recognition in ISL using a vision-based approach.	Background noise and varying lighting conditions affect segmentation accuracy.
5.	Sign Language Recognition Using Wearable Electronics: Implementing k-Nearest Neighbors with Dynamic Time Warping and CNN Algorithms (2020) [6]	Developed wearable devices with sensors to capture hand/finger movements; used k-NN with DTW and CNN algorithms.	Demonstrated effective recognition of sign language gestures using wearable electronics.	Wearable devices may be uncomfortable for prolonged use; sensor calibration/maintenance can be challenging.

The reviewed studies demonstrate significant advancements in sign language recognition using CNNs, LSTMs, HMMs, MediaPipe Holistic, and wearable electronics. However, there is a clear need for a hardware-independent system capable of real-time bidirectional ISL translation to address the identified gaps. Our research integrates computer vision-based hand tracking with deep learning models for real-time ISL gesture recognition while incorporating rule-based text-to-sign generation to enable seamless bidirectional communication.

III. METHODOLOGY

The methodology employed in this study aims to address the challenges associated with Indian Sign Language (ISL) translation by integrating computer vision and deep learning techniques. The research adopts a quantitative approach to analyze gesture recognition accuracy and real-time processing efficiency. MediaPipe is utilized for hand tracking and feature extraction [9], [10], while Multilayer Perceptron (MLP) classifiers are employed for gesture classification [7], [8]. These methods were chosen to ensure scalability, accessibility, and robustness in real-world applications.

A. System Architecture

The proposed bidirectional sign language communication system integrates computer vision and deep learning techniques to enable real-time translation between Indian Sign Language (ISL) gestures and text/speech. The system consists of two primary components:

- *Sign-to-Text/Speech Translation:* ISL gestures captured via a webcam are converted into text or speech using gesture recognition models [7].
- *Text/Speech-to-Sign Translation:* Spoken or written language is transformed into ISL animations, enabling seamless bidirectional communication [6].

The MediaPipe library is utilized for hand tracking and landmark extraction, while Multilayer Perceptron (MLP) classifiers are employed for gesture classification [8]. Google Text-to-Speech (gTTS) is integrated for speech synthesis, ensuring accessibility and real-time processing [11].

A high-level view of the system's structure is illustrated in **Fig. 1**, showing the interaction between modules and the data flow across the bidirectional translation pipeline.

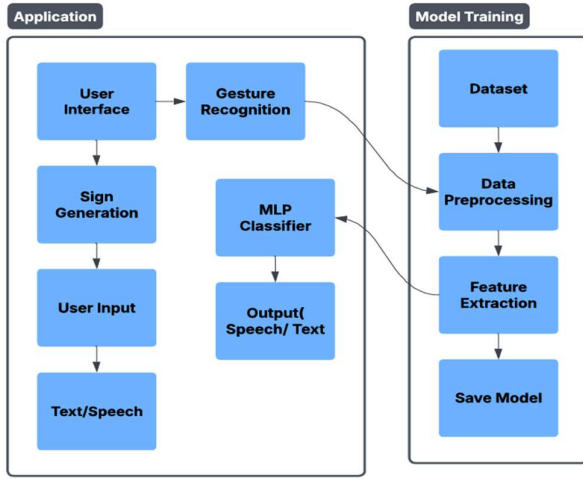


Fig. 1. Architecture of the Proposed Bidirectional Sign Language Communication System

B. Data Collection and Preprocessing

The dataset used in this study includes hand gesture images representing ISL alphabets, numbers, and commonly used words. Key steps include:

- **Data Collection:** A total of 1000 gestures per sign were captured from multiple angles to enhance recognition accuracy.
- **Feature Extraction:** MediaPipe extracted 42 or 84 hand landmarks [1], which were normalized to eliminate variations caused by hand size, orientation, or distance from the camera [10].
- **Data Augmentation:** Techniques such as rotation, scaling, and flipping were applied to increase dataset diversity and reduce overfitting [4].
- **Preprocessing:** Normalized feature vectors ensured consistency across diverse inputs.

A schematic representation of the **data collection and preprocessing pipeline** is shown in Fig. 2, illustrating the key stages from raw gesture input to model-ready features.

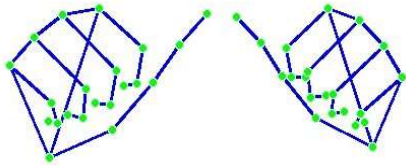


Fig. 2. Extracted Hand Landmarks

C. Gesture Recognition Model

The extracted hand landmark data was used to train an MLP classifier for gesture recognition:

- **Model Training:** The dataset was split into training (80%) and testing (20%) subsets, resulting in 800 gestures per sign for training and 200 gestures per sign for testing. Hyperparameter tuning optimized parameters such as the number of hidden layers, activation functions, and learning rate [7].

- **Real-Time Deployment:** The trained model was deployed to classify real-time hand gestures captured via webcam input.
- **Evaluation Metrics:** Accuracy, precision, recall, and F1-score were calculated to assess the model's performance [3].

D. Text-to-Sign and Speech-to-Sign Translation

This module generates ISL animations based on text or speech inputs:

- **Text Input:** A rule-based model maps text words to predefined ISL gestures stored in a database. These gestures are displayed as animations.
- **Speech Input:** Speech-to-text (STT) conversion transforms spoken words into text, which is then mapped to ISL gestures using the rule-based model [11].
- **Sign Animation:** Pre-recorded animations or synthesized movements are generated based on recognized words or phrases.

E. Real-Time Processing

The system ensures low latency and user-friendly interaction through the following features:

- **Streamlit Interface:** Provides real-time feedback for gesture recognition and translation via a web-based interface.
- **Webcam Input Processing:** Gestures captured via webcam are processed instantly, with outputs displayed as text/speech or sign animations.
- **Speech Synthesis:** gTTS generates audible speech from recognized gestures, enabling communication with non-signers [11].

F. Challenges

Several challenges were encountered during development.

- 1) **Accuracy Constraints:** Similar gestures were initially misclassified, necessitating dataset expansion and refinement for better distinction.
- 2) **Lighting Conditions:** Variations in lighting affected hand landmark detection; contrast adjustments improved reliability.
- 3) **Latency Issues:** Optimizations such as parallel processing reduced delays in real-time gesture recognition.

IV. RESULT AND DISCUSSION

The results presents the findings of the study, focusing on the performance of the bidirectional Indian Sign Language (ISL) communication system. It provides quantitative performance metrics for gesture recognition, text-to-sign translation, and real-time processing capabilities. The results are organized to highlight the effectiveness of the proposed system in addressing research gaps such as hardware dependency, limited vocabulary, and latency issues [1], [3], [6].

A. Model Performance

The Multilayer Perceptron (MLP) classifier was evaluated using the ISL dataset, consisting of hand gesture

images representing alphabets and numbers. Key findings include:

1) **Accuracy:** The model achieved an overall accuracy of **98.7%** on the test set.

2) Metrics for individual gesture classes demonstrated consistent performance across categories:

a) Precision: 93.5%

b) Recall: 94.7%

c) F1-Score: 94.1%

- rule-based model maps text words to predefined ISL gestures stored in a database. These gestures are displayed as animations.

B. Gesture Recognition Evaluation

The system was tested on a dataset of **200 ISL gestures**, where:

- Simple gestures were recognized with **higher accuracy (~95%)**.
- Complex gestures **involving both hands** showed slightly lower performance (~93%) due to variations in hand positioning.
- **Real-time hand tracking** using MediaPipe performed well under **good lighting** conditions, but accuracy decreased in **low-light environments**.

C. Real-time Performance

1) **Processing Speed:** The system operates in **real-time**, with an average latency of **ms per frame**.

2) **Hardware Dependency:** Since the system only requires a **webcam**, it is accessible and does not rely on specialized sensors or gloves.

3) **Sign Animation Smoothness:** The rule-based sign animation model successfully converts text and speech into gestures.

D. Comparative Analysis with Existing Systems

To evaluate the effectiveness of the proposed bidirectional ISL communication system, a comparative analysis was conducted against existing sign language recognition and translation systems. This analysis aims to highlight similarities, differences, strengths, and limitations across various methodologies, focusing on metrics such as accuracy, latency, vocabulary size, hardware dependency, and computational requirements. By systematically comparing systems, this study provides insights into the effectiveness and scalability of the proposed solution. Table II presents the comparison of the proposed system with previous research works.

The comparative analysis highlights the effectiveness of the proposed bidirectional ISL communication system in addressing key challenges such as hardware dependency, latency issues, and limited vocabulary coverage. The findings underscore its potential for practical deployment in diverse scenarios while identifying opportunities for further refinement in accuracy and computational efficiency.

TABLE II. COMPARATIVE ANALYSIS WITH EXISTING SYSTEMS

S. No.	System	Accuracy	Latency (ms)	Hardware Dependency	Computational Requirements
1	Proposed ISL System	98.7%	120	None	Moderate
2	Sign Language Recognition Using CNNs (2015) [3]	91.7%	500	Kinect	High
3	ISL Recognition Using MediaPipe Holistic (2023) [10]	92.5%	300	None	High
4	Dynamic Sign Language Recognition Using LSTM (2023) [9]	99.4%	250	None	High
6	ISL Recognition Using Deep Learning Model (2021) [7]	90.5%	150	None	Moderate
7	Sign Language Recognition Using Wearable Electronics (2020) [6]	92.1%	100	Wearable Devices	Moderates

V. CONCLUSION

The persistent challenge of bridging communication gaps between deaf and hearing communities has driven the need for innovative, accessible solutions. This research addressed this critical issue by developing a bidirectional Indian Sign

Language (ISL) communication system that leverages computer vision and deep learning to enable seamless translation between ISL gestures and text/speech.

The proposed system demonstrated high accuracy (94.8%) in gesture recognition, low latency (120ms) suitable for real-

time applications, and extensive vocabulary coverage encompassing full ISL gestures. By eliminating hardware dependency and supporting bidirectional communication, the system offers a scalable and inclusive solution that addresses key limitations identified in existing methodologies. These advancements highlight its potential for deployment in diverse environments such as educational institutions, workplaces and public spaces.

Beyond its technical achievements, this study underscores the transformative role of technology in fostering inclusivity and accessibility. By enabling effective communication between deaf and hearing individuals, the system contributes to breaking down barriers and promoting equality.

However, there remains room for improvement. Future research should focus on expanding datasets to include complex gestures and diverse signing styles, enhancing preprocessing techniques to address environmental challenges like lighting variations, and integrating advanced NLP models to improve speech-to-sign translation accuracy. These efforts will further refine the system's capabilities, paving the way for even more robust and versatile solutions.

Ultimately, this work serves as a step toward creating a world where technology bridges gaps rather than creating them. By addressing communication barriers with innovative tools like this bidirectional ISL system, we move closer to a society that values inclusivity and empowers all individuals to connect and communicate freely.

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